

GrapeLeaf AI

Petiole & Leaf Nutrient Analysis Report

Sample ID:	PDF-TEST-01	Crop:	Grape
Location:	Sangli	Season:	October
Report Date:	18 September 2026, 05:26	Standards:	October Pruning Reference

Sample Summary

Total Analyzed	Optimum	Safe Limit	Low	High	Above Safe Limit	Data Unavailable
5	1	1	1	1	1	11

Configured October Pruning Reference Standards

The following established values are applied for nutrient classification in this report. Reviewed and validated by viticulture specialists.

Nutrient Parameter	Unit	Reference Range / Safe Limit	Nutrient Parameter	Unit	Reference Range / Safe Limit
Nitrogen	%	1.31 – 2.4	Iron	ppm	40 – 80
Nitrate	ppm	700 – 1200	Manganese	ppm	40 – 100
Ammonium-N	ppm	500 – 800	Zinc	ppm	40 – 100
Phosphorus	%	0.41 – 0.8	Copper	ppm	5.01 – 10.0
Potassium	%	1.11 – 2.2	Boron	ppm	30 – 70
Calcium	%	0.74 – 1.14	Molybdenum	ppm	0.25 – 0.5
Magnesium	%	0.51 – 0.8	Sodium	%	< 0.5
Sulfur	%	0.14 – 0.27	Chloride	%	< 0.5

GrapeLeaf AI – Detailed Results

Nutrient Classification & Laboratory Comparison

Sample ID: PDF-TEST-01 | Crop: Grape | Season: October | Date: 18 September 2026, 05:26

Nutrient	Measured Value	Unit	Reference Range	Status	Analytical Explanation & Remarks
Nitrogen	1.1	%	1.31–2.4 %	Low	The entered value (1.1 %) is below the reference range of 1.31–2.4 %.
Nitrate	—	ppm	700–1200 ppm	Data Unavailable	No value was provided for this nutrient.
Ammonium-N	—	ppm	500–800 ppm	Data Unavailable	No value was provided for this nutrient.
Phosphorus	0.5	%	0.41–0.8 %	Optimum	The entered value (0.5 %) is within the reference range of 0.41–0.8 %.
Potassium	2.5	%	1.11–2.2 %	High	The entered value (2.5 %) is above the reference range of 1.11–2.2 %.
Calcium	—	%	0.74–1.14 %	Data Unavailable	No value was provided for this nutrient.
Magnesium	—	%	0.51–0.8 %	Data Unavailable	No value was provided for this nutrient.
Sulfur	—	%	0.14–0.27 %	Data Unavailable	No value was provided for this nutrient.
Iron	—	ppm	40–80 ppm	Data Unavailable	No value was provided for this nutrient.
Manganese	—	ppm	40–100 ppm	Data Unavailable	No value was provided for this nutrient.
Zinc	—	ppm	40–100 ppm	Data Unavailable	No value was provided for this nutrient.
Copper	—	ppm	5.01–10.0 ppm	Data Unavailable	No value was provided for this nutrient.
Boron	—	ppm	30–70 ppm	Data Unavailable	No value was provided for this nutrient.
Molybdenum	—	ppm	0.25–0.5 ppm	Data Unavailable	No value was provided for this nutrient.
Sodium	0.6	%	Less than 0.5 %	Above Safe Limit	The entered value (0.6 %) meets or exceeds the safe limit of 0.5 %. Elevated levels may indicate salinity concerns.
Chloride	0.2	%	Less than 0.5 %	Safe	The entered value (0.2 %) is below the safe limit of 0.5 %.

GrapeLeaf AI – Observations & Guidance

Nutrient Interpretation, Agronomic Next Steps & Safeguards

[LOW NUTRIENT OBSERVATIONS]

Nitrogen (1.1 %) — The entered value (1.1 %) is below the reference range of 1.31–2.4 %. *Guidance: Review soil conditions, crop growth stage, and fertilizer history. Consult a qualified agricultural professional before making any amendments. Consider verification through appropriate laboratory re-testing.*

[HIGH NUTRIENT OBSERVATIONS]

Potassium (2.5 %) — The entered value (2.5 %) is above the reference range of 1.11–2.2 %. *Guidance: Avoid unnecessary additional application of this nutrient until the result is verified. Review fertilizer history and consider re-testing to confirm. Consult an agricultural professional.*

[ABOVE SAFE LIMIT OBSERVATIONS (SALINITY STRESS)]

Sodium (0.6 %) — The entered value (0.6 %) meets or exceeds the safe limit of 0.5 %. Elevated levels may indicate salinity concerns. *Guidance: Consult a qualified agricultural professional for further assessment. Consider soil testing and irrigation-water testing to evaluate salinity conditions. Review current irrigation sources and soil amendment practices.*

[OPTIMUM & SAFE STATUS]

The following 2 nutrient(s) meet October Pruning reference standards: **Phosphorus (0.5 %), Chloride (0.2 %)**. Continue balanced irrigation and maintenance fertilization.

[DATA UNAVAILABLE]

No measured values were provided for: Nitrate, Ammonium-N, Calcium, Magnesium, Sulfur, Iron, Manganese, Zinc, Copper, Boron, Molybdenum. These were excluded from classification.

Recommended Agronomic Next Steps

1. **Review with a specialist:** Consult a certified viticulturist or local agricultural extension officer prior to adjusting fertigation schedules.
2. **Salinity screening:** If Sodium or Chloride is above safe limits, test soil electrical conductivity (EC) and irrigation water source.
3. **Growth stage confirmation:** Re-verify tissue samples at berry set and veraison to track seasonal mobilization curves.
4. **Maintain application logs:** Keep detailed logs of all soil and foliar applications to diagnose potential ion antagonisms.

Precautions & Biological Limitations

- **Standards Scope:** Calibrated for grape petioles at October forward pruning. Not applicable to other phenological stages.
- **Varietal Differences:** Table varieties (Thompson Seedless) and wine varieties have distinct nutrient demands.
- **Rootstock Influence:** Rootstocks like Dogridge or 110R alter potassium, magnesium, and chloride uptake significantly.
- **Lab Variation:** Differences in petiole washing, drying, and digestion techniques can introduce measurement variance.

***Educational Decision Support Disclaimer:** This report provides educational decision support based strictly on entered values and configured October Pruning reference standards. It is not a substitute for qualified agricultural advice. Do not apply chemical fertilizers or amendments without professional verification. GrapeLeaf AI does not guarantee yield improvement, disease detection, or crop recovery.*